



Supplementary Materials for

An expert-level generalist AI for abdominal CT diagnosis

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Tables S1 to S13
MDAR Reproducibility Checklist

Materials and Methods

Evaluation of anatomical perception module

We evaluated the module's segmentation performance on the TotalSegmentator test set, achieving a mean Dice score of 0.917 (95% CI, 0.894–0.937) across the 18 target structures (fig. S11), compared to 0.928 (95% CI, 0.909–0.947) from the publicly released TotalSegmentator model ($P = 0.17$). The pretrained weights of both the vision encoder and the anatomical perception module were then used to initialize the RADAR framework. Additionally, we applied the public TotalSegmentator model to the entire training set to generate pseudo-labels for target anatomies. These pseudo-labels were incorporated into later training stages to further optimize the segmentation objective and preserve RADAR's anatomical perception capability.

Ablation study

We provided a comprehensive ablation analysis of the RADAR framework. First, we investigated the impact of the backbone architecture and the anatomical perception module on the model's performance.

CNN vs. ViT encoder. We compared the performance of a CNN-based encoder with a Vision Transformer (ViT). Specifically, RADAR with the CNN encoder achieved an AUC of 0.903 (95% CI, 0.898–0.907), surpassing the ViT's performance of 0.755 (95% CI, 0.749–0.761) ($P < 0.001$; fig. S12). We attribute the ViT's underperformance to the inherent requirements of 3D CT interpretation, which demands high-resolution local perception to detect subtle abnormalities. CNN architectures, such as nnU-Net, are inherently optimized for modeling these fine-grained spatial features, whereas the global attention mechanism of Transformers may overlook the localized nuances critical for accurate diagnosis.

Effectiveness of anatomical perception module. RADAR demonstrates significantly enhanced performance with the integration of the anatomical perception module. Excluding this module from RADAR led to a notable AUC drop from 0.903 (95% CI, 0.898–0.907) to 0.883 (95% CI, 0.876–0.891) ($P < 0.001$; fig. S12). The observed decline substantiates that capturing spatial anatomical priors is crucial for enriching the semantic representation of anatomy-level features.

Second, to further understand the efficacy of our proposed anatomical contrastive learning strategy, we analyzed several key design choices within the anatomy-level alignment framework.

Whole-image vs. anatomical image–text alignment. Compared to the traditional whole-image image-text alignment approach, our anatomical alignment strategy in RADAR demonstrated clear superiority. Replacing anatomical alignment with a whole-image baseline in RADAR significantly degraded diagnostic performance, as evidenced by the sharp decline in AUC from 0.903 (95% CI, 0.898–0.907) to 0.673 (95% CI, 0.665–0.687) ($P < 0.001$; fig. S12). This underscores that whole-image alignment fails to capture the fine-grained, localized structure required for feature alignment, whereas our anatomical alignment effectively bridges the gap between local visual features and specific medical descriptions.

Vanilla InfoNCE loss vs. adaptive contrastive loss. Notably, the standard InfoNCE loss failed to converge in our dataset, resulting in severe numerical instability and training collapse (nan values; fig. S12, fifth column). In contrast, RADAR with adaptive contrastive loss yielded a robust AUC of 0.903 (95% CI, 0.898–0.907). This comparison underscores the indispensable role of the adaptive contrastive mechanism in mitigating the impact of false negatives. In the context of clinical diagnosis, semantically similar samples from both normal and abnormal classes are often incorrectly treated as negatives in vanilla InfoNCE. This misclassification

severely disrupts the semantic alignment between images and text. Our approach effectively addresses this issue and ensures stable convergence.

Anatomical feature aggregation: average pooling vs. learnable query tokens. We evaluated different strategies for aggregating anatomical features. Compared to simple average feature pooling, the use of learnable query tokens in RADAR achieved a significantly higher AUC, with a performance gain of 0.009 ($P < 0.01$; fig. S12). This improvement suggests that learnable queries can effectively aggregate anatomy-specific features to yield highly discriminative representations.

Momentum-text similarity weighting. We finally assessed the necessity of the momentum-text similarity weighting mechanism. The removal of this component in RADAR caused the performance to decay significantly, with the AUC dropping from 0.903 (95% CI, 0.898–0.907) to 0.891 (95% CI, 0.885–0.897) ($P < 0.001$; fig. S12). This statistically significant reduction indicates that the weighting scheme contributes positively to feature alignment.

Network architecture

RADAR consists of two coordinated branches, namely a vision branch and a text branch, which cooperate to learn fine-grained alignment between anatomical structures in CT volumes and corresponding descriptions in radiology reports (fig. S9). We denote a mini-batch of data as $\{I_i, R_i\}_{i=1}^B$, where B is the number of samples in the batch, I_i and R_i represents the CT volume and its corresponding radiology report of the i -th patient. Each sample was decomposed into K anatomical structures, denoted by $\{I_{i,k}, R_{i,k}\}_{k=1}^K$. The vision branch incorporates a 3D U-Net encoder to extract dense whole-volume feature map F_i and a segmentation decoder that serves as the anatomical perception module, enabling spatial localization of multiple anatomies within the scan. Both the encoder and decoder are automatically configured by the nnU-Net toolbox (32) based on the voxel resolution, volume

size of the whole training data. For each class of anatomical structure, indexed by k , the corresponding segmentation mask $M_{i,k}$ predicted by the decoder serves as a spatial reference to isolate the foreground region in the encoder’s multi-scale feature maps. Specifically, all spatial positions within the mask are indexed, and the feature vectors at these positions are extracted and stacked to form the anatomy-specific feature set $F_{i,k} \in \mathbb{R}^{S \times d}$, where S denotes the number of foreground positions in $M_{i,k}$ and d is the dimension of the feature vectors. And each row of $F_{i,k}$ corresponds to one spatial location’s feature representation of the k -th anatomy. A learnable query token $q_{i,k} \in \mathbb{R}^{1 \times d}$ is initialized and updated through a cross-attention mechanism, where the query attends to the set of Key/Value tokens derived from $F_{i,k}$. The aggregation is formalized as

$$\tilde{q}_{i,k} = \text{Softmax} \left(\frac{q_{i,k} W_Q (F_{i,k} W_K)^T}{\sqrt{d}} \right) (F_{i,k} W_V)$$

where W_Q , W_K , W_V are projection matrices and the resulting $\tilde{q}_{i,k}$ is processed by a linear projection to yield the visual CLS token $V_{i,k}$ for the k -th anatomy. The text branch comprises a word tokenization stage, followed by a BERT-base encoder (58) (12 Transformer layers, 768 hidden dimensions, 12 attention heads) that processes anatomy-specific report fragments $R_{i,k}$ to produce text CLS tokens $T_{i,k}$ from the CLS output embedding.

Anatomical image-text alignment

To mitigate alignment bias stemming from heterogeneous anatomical complexity, RADAR performs anatomy-level image–text alignment via fine-grained contrastive learning. Given a mini-batch, only anatomies that remain structurally complete after random cropping of the input volumes (patch size $96 \times 256 \times 384$) are considered for loss computation. For each class of anatomical structure, indexed by k , the probability of matching a visual CLS token from the i -th sample to a textual CLS token from j -th sample is given by

$$p_{i,k,j}^{i2t} = \frac{\exp(\langle V_{i,k}, T_{j,k} \rangle / \tau)}{\sum_{k'=1}^{N_k} \exp(\langle V_{i,k}, T_{k',k} \rangle / \tau)}, \quad p_{i,k,j}^{t2i} = \frac{\exp(\langle T_{i,k}, V_{j,k} \rangle / \tau)}{\sum_{k'=1}^{N_k} \exp(\langle T_{i,k}, V_{k',k} \rangle / \tau)}$$

where $\langle \cdot, \cdot \rangle$ denotes cosine similarity, τ is a learnable temperature parameter, and N_k is the number of complete samples for the k -th anatomy in the batch. The image–text contrastive loss aggregates bidirectional alignment, defined as

$$L_{itc} = -\frac{1}{2KN_k^2} \sum_{k=1}^K \sum_{i=1}^{N_k} \sum_{j=1}^{N_k} [y_{i,k,j}^{i2t} \log p_{i,k,j}^{i2t} + y_{i,k,j}^{t2i} \log p_{i,k,j}^{t2i}]$$

where $y_{i,k,j}^{i2t}$ and $y_{i,k,j}^{t2i}$ denote anatomy-specific target labels used for optimization, which will be introduced in the next subsection. Segmentation and alignment objectives are jointly optimized according to $\mathcal{L} = \lambda_{itc} \mathcal{L}_{itc} + \lambda_{seg} \mathcal{L}_{seg}$, with $\lambda_{itc}, \lambda_{seg}$ set to 1.0.

Adaptive contrastive modeling

Beyond standard instance-level pairing, RADAR uses adaptive contrastive learning to reduce false negatives caused by semantic similarities among samples from different patients. The core of this strategy is the construction of an adaptive label matrix y that unifies default identity matches, corrections for normal pairs, and similarity-based soft labels for abnormal pairs. For normal cases, anatomy-level binary labels are extracted from reports by the Qwen language model (57). Pairs of visual and textual CLS tokens from two normal samples of the same anatomy are reassigned positive labels to encourage feature proximity. For abnormal cases, a momentum-updated text encoder (EMA = 0.995) computes cosine similarities between abnormal anatomy descriptions, which are used as soft labels to partially align semantically similar pathologies even across different patients. Let $i, j \in 1, \dots, N_k$ denote visual–text instances within a given anatomy indexed by k , the adaptive label matrix is defined as

$$y_{i,k,j} = \begin{cases} 1, & i = j \\ 1, & (i,j) \text{ are normal samples of the } k\text{-th anatomy} \\ s_{i,k,j}, & (i,j) \text{ are abnormal samples of the } k\text{-th anatomy} \\ 0, & \text{otherwise} \end{cases}$$

where $s_{i,k,j}$ denotes the normalized similarity scores between abnormal pairs, computed from the momentum-updated text encoder. We then perform row-wise L1 normalization $y_{i,k,j} \leftarrow \frac{y_{i,k,j}}{\sum_{j'=1}^{N_k} |y_{i,k,j'}|}$, ensuring that each row forms a valid probability distribution over all text instances for a given visual query. The resulting label matrix is directly integrated into the image–text contrastive loss $\mathcal{L}_{itc}$, enabling RADAR to learn anatomically aware and diagnostically robust visual–language representations.

Data-driven prompt ensemble

In addition to rule-based prompt construction, we introduce a data-driven prompt ensemble strategy to improve robustness. For each finding f , we first identify positive and negative cases in the training set based on reports. From the positive reports, we extract descriptions and conclusions directly related to f to form the positive prompt set $\mathcal{P}_f^+ = \{p_{f,1}^+, p_{f,2}^+, \dots, p_{f,n}^+\}$. Similarly, from the negative reports, we extract expressions explicitly indicating the absence of f or denoting the normal/healthy status of the corresponding anatomy, yielding the negative prompt set $\mathcal{P}_f^- = \{p_{f,1}^-, p_{f,2}^-, \dots, p_{f,n}^-\}$. Each prompt is encoded by the text encoder $E_t(\cdot)$ to obtain:

$$T_{f,i}^+ = E_t(p_{f,i}^+), \quad T_{f,i}^- = E_t(p_{f,i}^-), \quad i = 1, \dots, n,$$

where n is the number of ensemble prompts, and is empirically set to 500 (fig. S13). We then aggregate the embeddings of all prompts in each set via mean pooling to form an ensemble representation:

$$\bar{T}_f^+ = \frac{1}{n} \sum_{i=1}^n T_{f,i}^+, \quad \bar{T}_f^- = \frac{1}{n} \sum_{i=1}^n T_{f,i}^-.$$

At inference time, given a visual embedding V_a for the anatomy a where the finding f may reside, we compute cosine similarities:

$$s_f^+ = \langle Va, \bar{\mathbf{T}}_f^+ \rangle, \quad s_f^- = \langle Va, \bar{\mathbf{T}}_f^- \rangle,$$

The final probability of f being present is obtained via a softmax over the positive and negative similarities:

$$p_f = \frac{\exp(s_f^+/\tau)}{\exp(s_f^+/\tau) + \exp(s_f^-/\tau)},$$

where τ is a learnable temperature parameter. This ensemble approach leverages diverse prompts extracted from real clinical language, reducing overfitting to a single phrasing and improving robustness to variations in report expression. We empirically set the number of ensembled prompts to 500 (fig. S13).

RADAR+

To extend RADAR’s applicability beyond organ-specific tasks, we introduce a simple yet efficient global image-text alignment module into RADAR+, designed to align whole-volume CT representations with entire radiology reports. In the vision branch, the visual encoder produces a feature map F_i for the i -th CT volume. A learnable global query token $\tilde{q}_i^{global} \in \mathbb{R}^{1 \times d}$ is initialized and updated via a cross-attention mechanism, where the query attends to all spatial positions of the feature map:

$$\tilde{q}_i^{global} = \text{Softmax} \left(\frac{q_i^{global} W_Q (F_i W_K)^\top}{\sqrt{d}} \right) (F_i W_V),$$

where W_Q , W_K , W_V are projection matrices. The resulting $\tilde{q}_i^{global}$ is processed by a linear projection to yield the global visual CLS token $\mathbf{V}_i^{global}$. In the text branch, the complete patient report R_i is tokenized and passed through the text encoder to produce a global text CLS token $\mathbf{T}_i^{global} \in \mathbb{R}^d$. The probability of matching a global visual CLS token from the i -th sample to a global text CLS token from the j -th sample is given by:

$$p_{\text{global}}^{i2t}(i, j) = \frac{\exp(\langle \mathbf{V}_i^{\text{global}}, \mathbf{T}_j^{\text{global}} \rangle / \tau)}{\sum_{m=1}^B \exp(\langle \mathbf{V}_i^{\text{global}}, \mathbf{T}_m^{\text{global}} \rangle / \tau)},$$

$$p_{\text{global}}^{t2i}(i, j) = \frac{\exp(\langle \mathbf{T}_i^{\text{global}}, \mathbf{V}_j^{\text{global}} \rangle / \tau)}{\sum_{m=1}^B \exp(\langle \mathbf{T}_i^{\text{global}}, \mathbf{V}_m^{\text{global}} \rangle / \tau)},$$

where $\langle \cdot, \cdot \rangle$ denotes cosine similarity, B is the batch size, and τ is a learnable temperature parameter. The global image-text contrastive loss aggregates bidirectional alignment:

$$\mathcal{L}_{\text{global}} = -\frac{1}{2B^2} \sum_{i=1}^B \sum_{j=1}^B [y_{\text{global}}^{i2t}(i, j) \log p_{\text{global}}^{i2t}(i, j) + y_{\text{global}}^{t2i}(i, j) \log p_{\text{global}}^{t2i}(i, j)]$$

where $y_{\text{global}}^{i2t}(i, j)$ and $y_{\text{global}}^{t2i}(i, j)$ are global-level labels (1 if i and j are the same case, otherwise 0).

We employ an interleaved training strategy in which mini-batches are alternately processed by the anatomical alignment and the global alignment within each training epoch. For a global-alignment mini-batch, the segmentation loss $\mathcal{L}_{\text{seg}}$ and the global alignment loss $\mathcal{L}_{\text{global}}$ are computed, whereas for an anatomical-alignment mini-batch, $\mathcal{L}_{\text{seg}}$ and the anatomy-level contrastive loss $\mathcal{L}_{\text{itc}}$ are computed. This interleaved optimization preserves anatomy-specific discriminative ability through organ-level alignment while, at the same time, enabling the model to acquire whole-body diagnostic capability via global alignment.

Findings prompt alignment

While the original RADAR training objective focuses on anatomy-level image–text alignment, such alignment primarily emphasizes general anatomical semantics, which may not fully encompass certain detailed findings described in radiology reports. To further enhance RADAR’s capability to capture and discriminate these explicit findings, we introduce a findings prompt alignment approach and perform fine-tuning on RADAR. This fine-tuning aims to strengthen the correspondence between visual representations of anatomical regions

and textual prompts that explicitly state the presence or absence of clinically significant findings.

Findings extraction. For each anatomy-specific report fragment $R_{i,k}$ from the i -th sample and anatomical structure k , the Qwen LLM was used to extract the set of clinical findings relevant to that anatomy $\mathcal{F}_{i,k} = \{f_{i,k}^{(m)}\}_{m=1}^{M_{i,k}}$, where $M_{i,k}$ denotes the number of unique findings detected in $R_{i,k}$.

Prompt generation. For each finding $f_{i,k}^{(m)}$, we construct both a positive and a negative textual prompt:

$$P_{i,k,m}^+ = "\{finding\}", \quad P_{i,k,m}^- = "no \{finding\}"$$

These prompts are tokenized and encoded with the text branch's BERT-base encoder, yielding the corresponding CLS embeddings:

$$\mathbf{T}_{i,k,m}^+, \quad \mathbf{T}_{i,k,m}^- \in \mathbb{R}^d$$

The embeddings are L2-normalized prior to similarity computation.

Similarity and probability computation. Given the visual CLS token $\mathbf{V}_{i,k} \in \mathbb{R}^d$ for anatomy k in sample i , we compute cosine similarities:

$$s_{i,k,m}^+ = \langle \mathbf{V}_{i,k}, \mathbf{T}_{i,k,m}^+ \rangle, \quad s_{i,k,m}^- = \langle \mathbf{V}_{i,k}, \mathbf{T}_{i,k,m}^- \rangle,$$

where $\langle \cdot, \cdot \rangle$ denotes cosine similarity. The similarities are converted into probabilities via a softmax over the positive/negative pair:

$$p_{i,k,m}^+ = \frac{\exp(s_{i,k,m}^+/\tau_f)}{\exp(s_{i,k,m}^+/\tau_f) + \exp(s_{i,k,m}^-/\tau)},$$

$$p_{i,k,m}^- = \frac{\exp(s_{i,k,m}^-/\tau_f)}{\exp(s_{i,k,m}^+/\tau_f) + \exp(s_{i,k,m}^-/\tau)}$$

where τ is a learnable temperature parameter.

Loss function. The LLM-extracted finding label $y_{i,k,m} \in \{0,1\}$ indicates presence (1) or absence (0) of the finding m . We define the findings prompt fine-tuning loss as:

$$\mathcal{L}_{fpa} = -\frac{1}{\sum_{i,k} M_{i,k}} \sum_{k=1}^K \sum_{i=1}^{N_k} \sum_{m=1}^{M_{i,k}} [y_{i,k,m} \log p_{i,k,m}^+ + (1 - y_{i,k,m}) \log p_{i,k,m}^-]$$

The overall training objective is:

$$\mathcal{L} = \mathcal{L}_{itc} + \mathcal{L}_{seg} + \lambda_{fpa} \mathcal{L}_{fpa}$$

where λ_{fpa} controls the contribution of the findings prompt alignment term, and is set to 1 by default.

Performance evaluation. We compared the performance of RADAR before and after fine-tuning on both internal and external multicenter test cohorts (fig. S7). In the internal test cohort, the AUC increased significantly from 0.913 (95% CI, 0.911–0.915) to 0.940 (95% CI, 0.937–0.944) ($P < 0.001$). In the external multicenter test cohort, the AUC increased significantly from 0.895 (95% CI, 0.889–0.900) to 0.927 (95% CI, 0.925–0.928) ($P < 0.001$). These results indicate that the proposed fine-tuning strategy can effectively further enhance RADAR’s performance.

Computational complexity and inference strategy

We adopt a sliding-window inference strategy rather than global-window (full-volume) inference. This choice, inspired by established practice in 3D medical image segmentation, aims to keep the inference input size consistent with that used in training, thereby improving model compatibility, while also being more memory-efficient in limited-resource environments by processing smaller sub-volumes sequentially. We compared the two strategies in terms of AUC, speed, and memory consumption in table S13. Sliding-window inference achieves an AUC of 0.913 (95% CI, 0.911–0.915), slightly higher than the 0.909 (95% CI, 0.907–0.912) of full-volume inference, likely because it preserves the same input size as in training. In terms of runtime on an NVIDIA H20 GPU, the average inference time across all cases in the test set is approximately 1.3 seconds per CT scan for sliding-window inference,

compared to only 0.6 seconds for full-volume inference, as the latter avoids multiple sequential passes. However, this speed advantage comes at a significant memory cost: full-volume inference incurs a peak GPU memory consumption of 70 GB, compared to 21 GB for the sliding-window approach, which poses a challenge for practical deployment.

Regarding training cost, using $24 \times$ NVIDIA H20 GPUs, training RADAR for 30 epochs on 424,911 examinations required 213 hours of wall-clock time. Given an input size of $96 \times 256 \times 384$, the computational complexity of RADAR is 5.54 TFLOPs.

Balancing performance, efficiency, and hardware constraints, we therefore recommend the sliding-window strategy as the final choice for deployment, as it offers slightly better performance while remaining far more memory-friendly.

Supplementary Figures and Tables

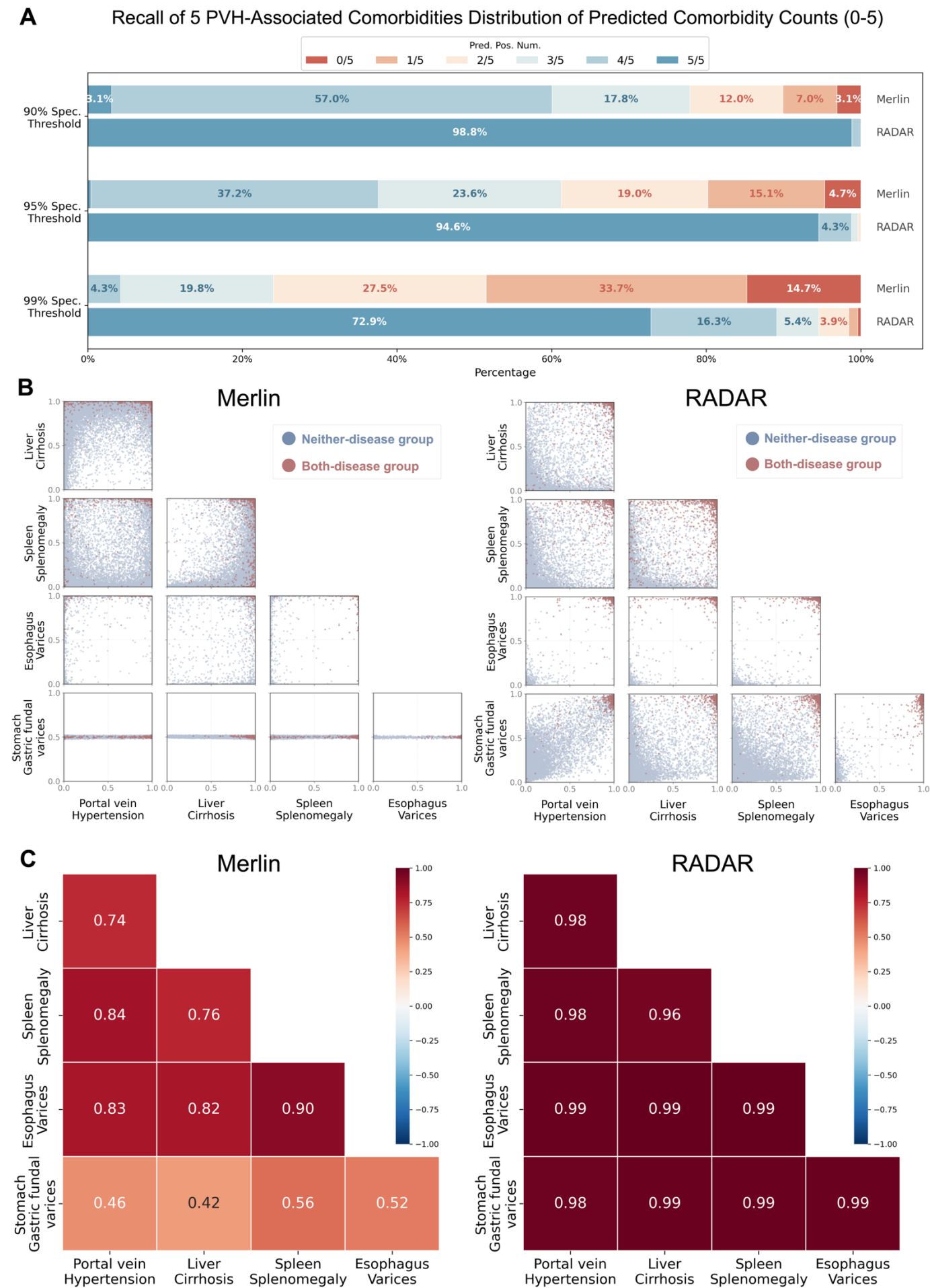


Figure S1. Comparative analysis of Merlin and RADAR in identifying five portal vein hypertension (PVH)-associated comorbidities. (A) Comparison of recall performance for five PVH-associated comorbidities, illustrating the distribution of correctly identified conditions (0–5) across three specificity thresholds (90%, 95%, 99%) in a test cohort of 258 patients with confirmed multi-morbidity. (B) Pairwise diagnostic score distributions demonstrating the models' discriminative capability between neither-disease and both-disease groups. (C) Consistency analysis of model diagnostic scoring. Heatmaps quantify the model's propensity for aligned risk predictions across comorbidity pairs, calculated over 5,000 bootstrap iterations. Each iteration involves paired sampling from neither-disease and both-disease groups, with consistency scores reflecting the frequency of synchronized score shifts: +1 for simultaneous elevation, -1 for simultaneous decrease, and 0 for inconsistent trends.

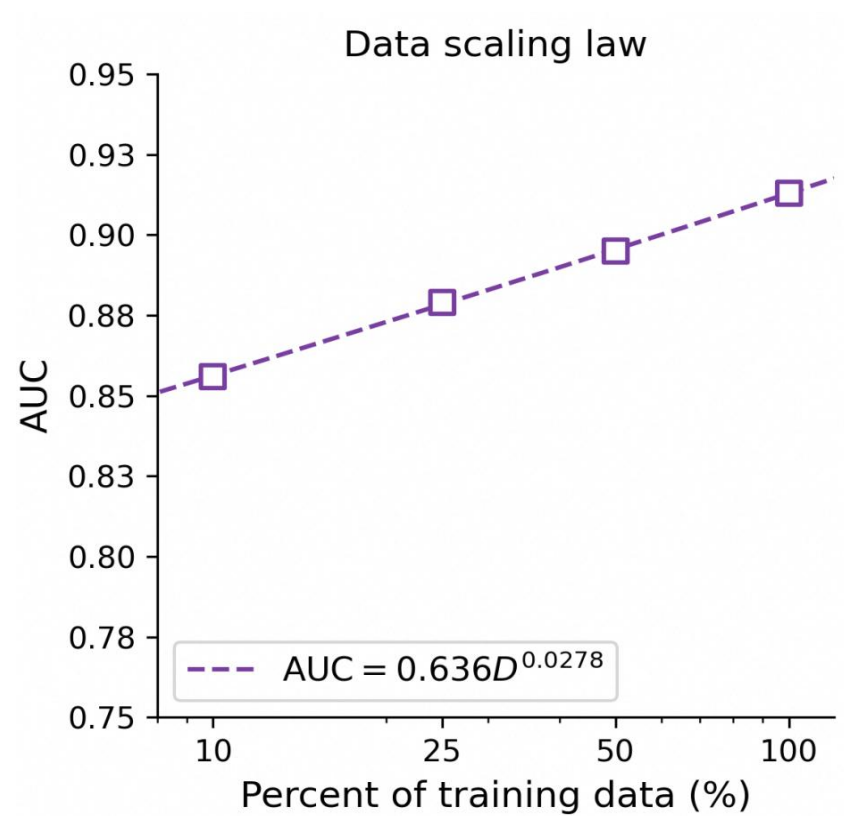


Figure S2. Data scaling law. The X-axis represents the proportion of training data used, and the Y-axis shows the mean AUC of the model on the internal testing cohort. The fitted curve indicates that RADAR continues to achieve performance gains when increasing training data.

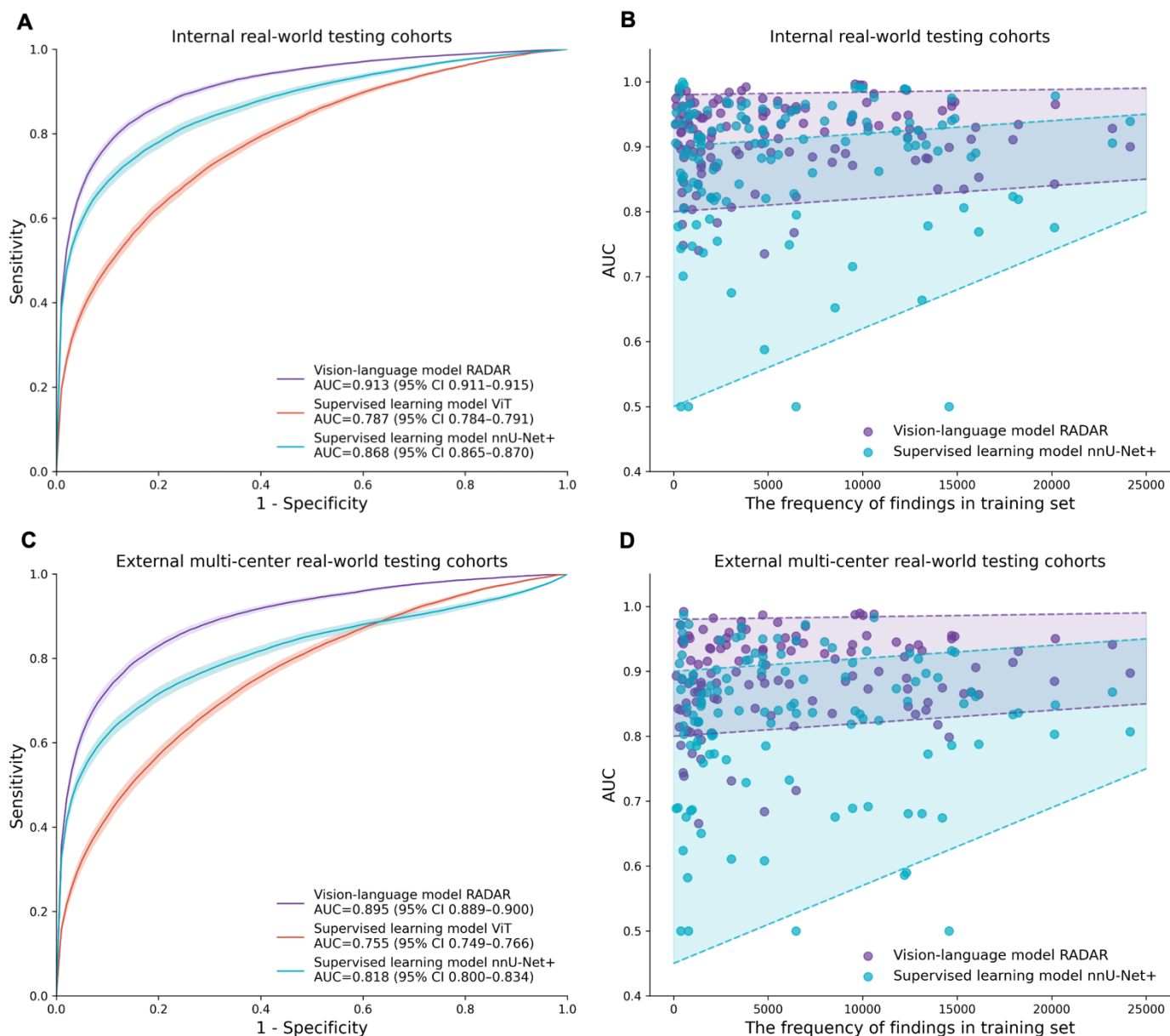
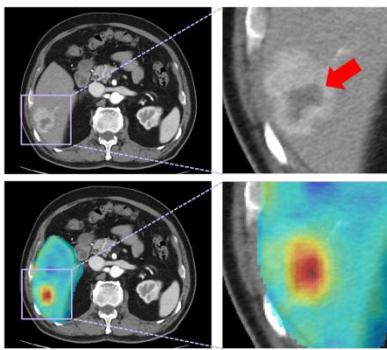


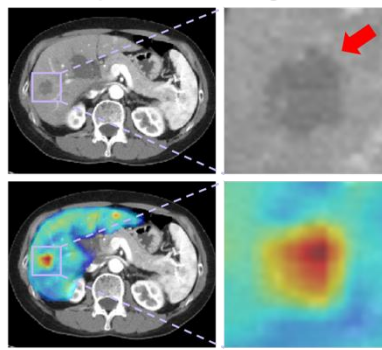
Figure S3. Comparison of RADAR and the supervised learning model ViT and nnU-Net+. (A) and (C) ROC curves for the internal real-world testing cohort (A) and the external multicenter real-world testing cohort (C). The colored bands indicate the standard error of the ROC curves, with each band's central line representing the mean AUC calculated across multiple findings. Compared with RADAR, both supervised models showed a larger decrease in AUC from the internal to the external cohort, indicating greater performance degradation in the external validation. (B) and (D) Relationship between training-set finding frequency and AUC in the internal (B) and external (D) testing cohorts. RADAR

demonstrates robust performance across finding frequencies, whereas nnU-Net+ declined for low-frequency findings.

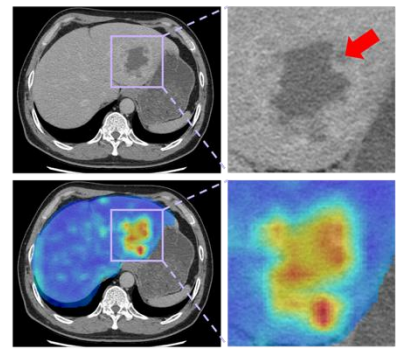
Hepatocellular carcinoma



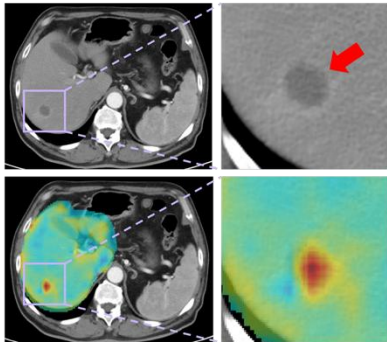
Hepatic hemangioma



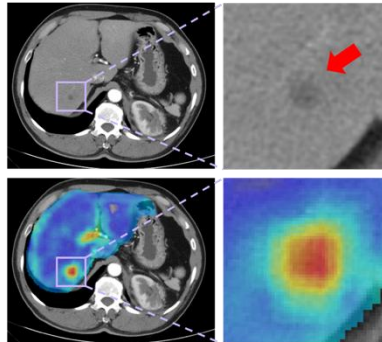
Hepatic abscess



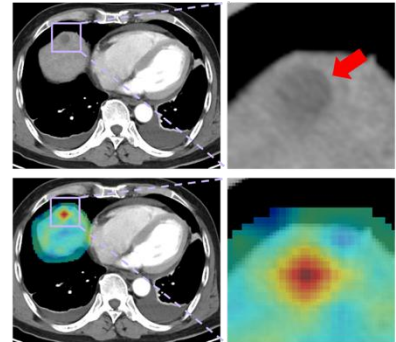
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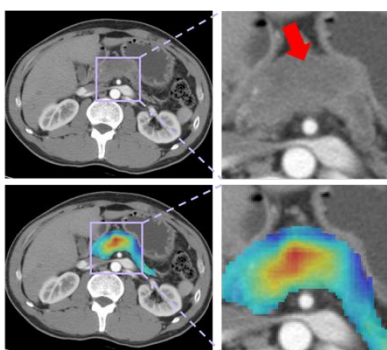
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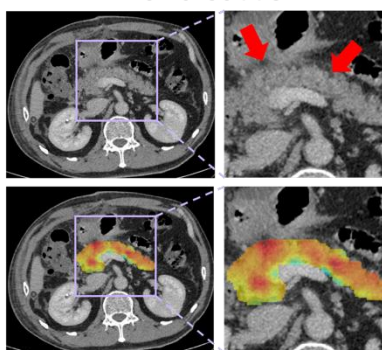
Intrahepatic cholangiocarcinoma



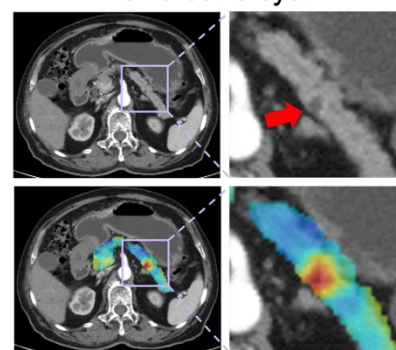
Pancreatic cancer



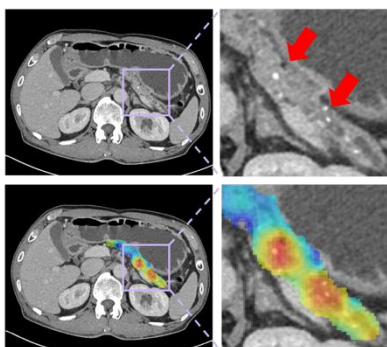
Pancreatitis



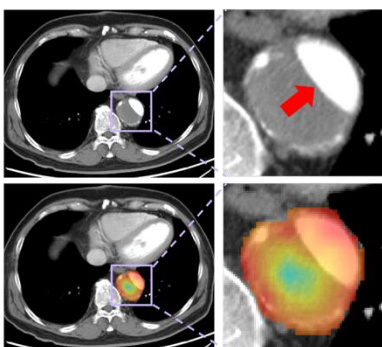
Pancreatic cyst



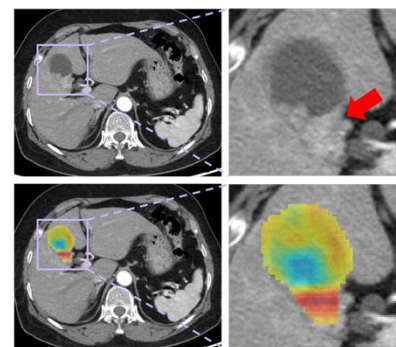
Pancreatic duct calculus



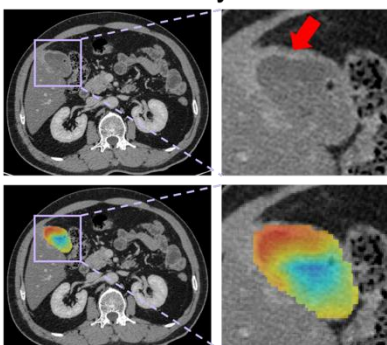
Aortic dissection



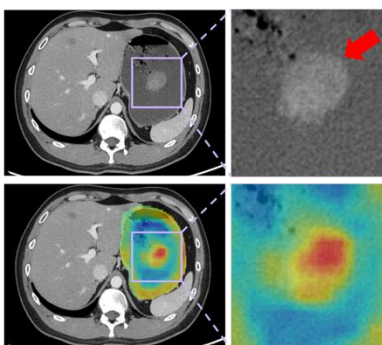
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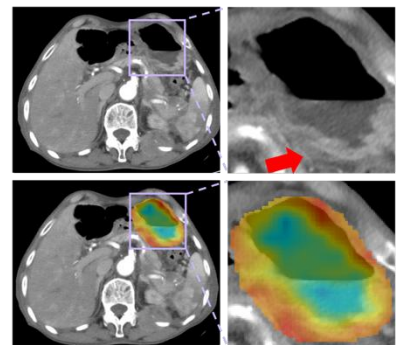
Cholecystitis



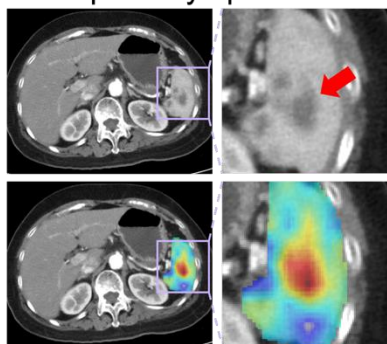
Gastrointestinal stromal tumor



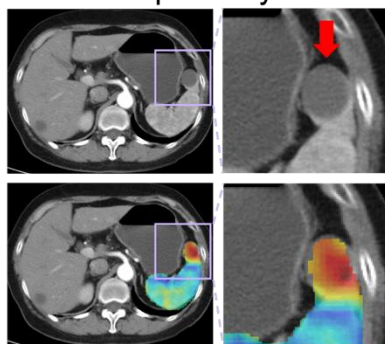
Gastric cancer



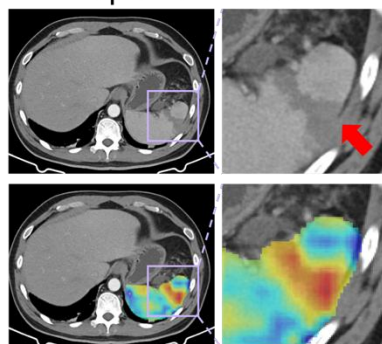
Splenic lymphoma



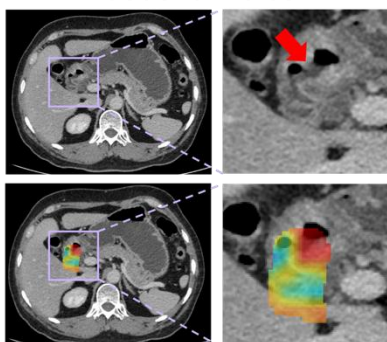
Splenic cyst



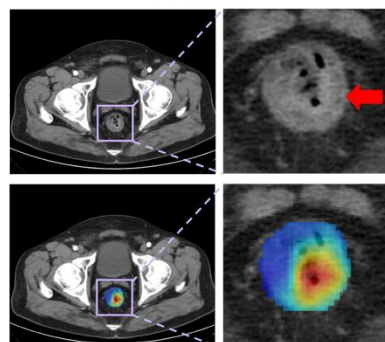
Splenic infarction



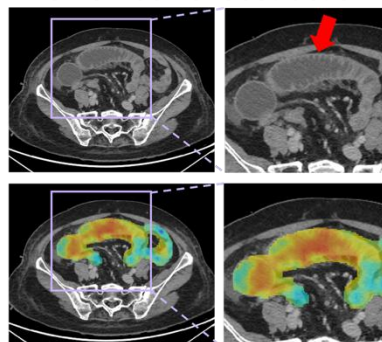
Duodenum ulcer



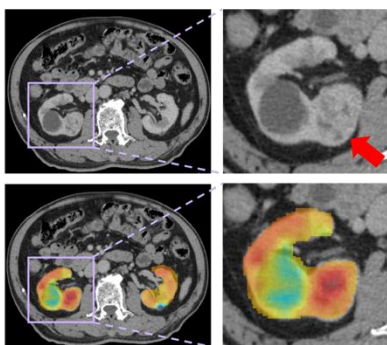
Rectal cancer



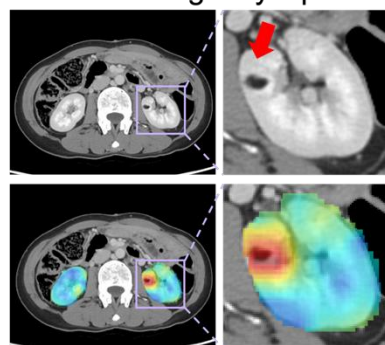
Small-bowel obstruction



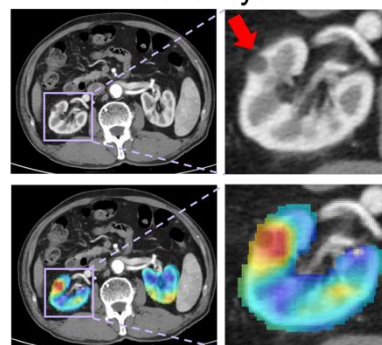
Renal cell carcinoma



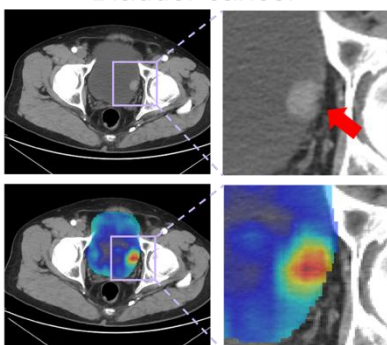
Renal angiomyolipoma



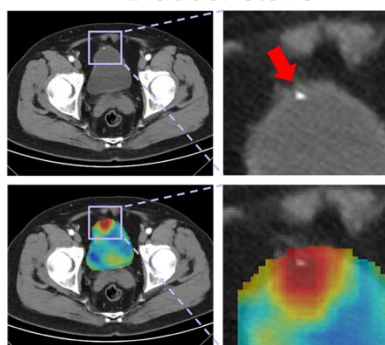
Renal cyst



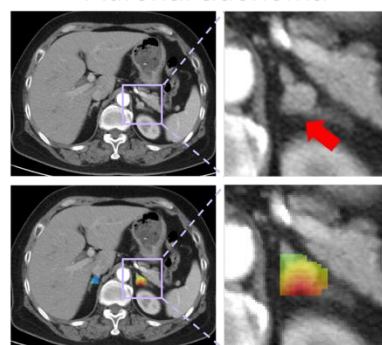
Bladder cancer



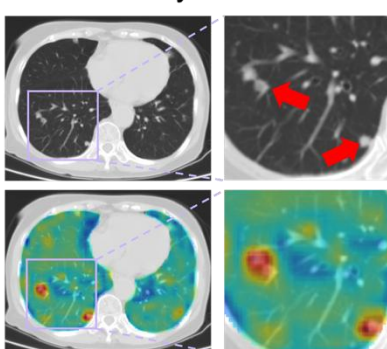
Bladder stone



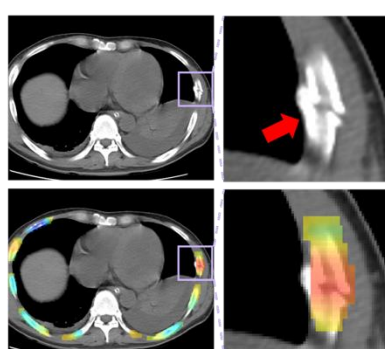
Adrenal adenoma



Pulmonary metastasis



Rib fracture



Rib metastasis

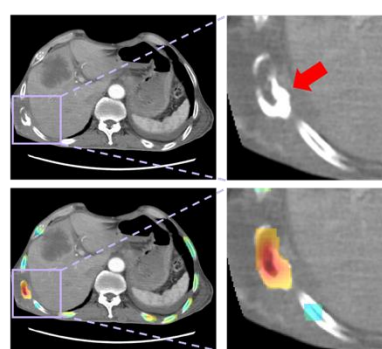


Figure S4. Representative attention maps of 30 diseases across 14 organs. Each panel is arranged in a 2×2 grid: top-left, the axial abdominal contrast-enhanced CT slice; bottom-left, the same slice with the RADAR attention map overlaid; top-right, a magnified view of the boxed lesion region from the original image; bottom-right, a magnified view of the attention-overlay image. Attention maps were generated by Grad-CAM, warmer colors (blue to red) indicate regions of greater model attention and influence on its predictions.

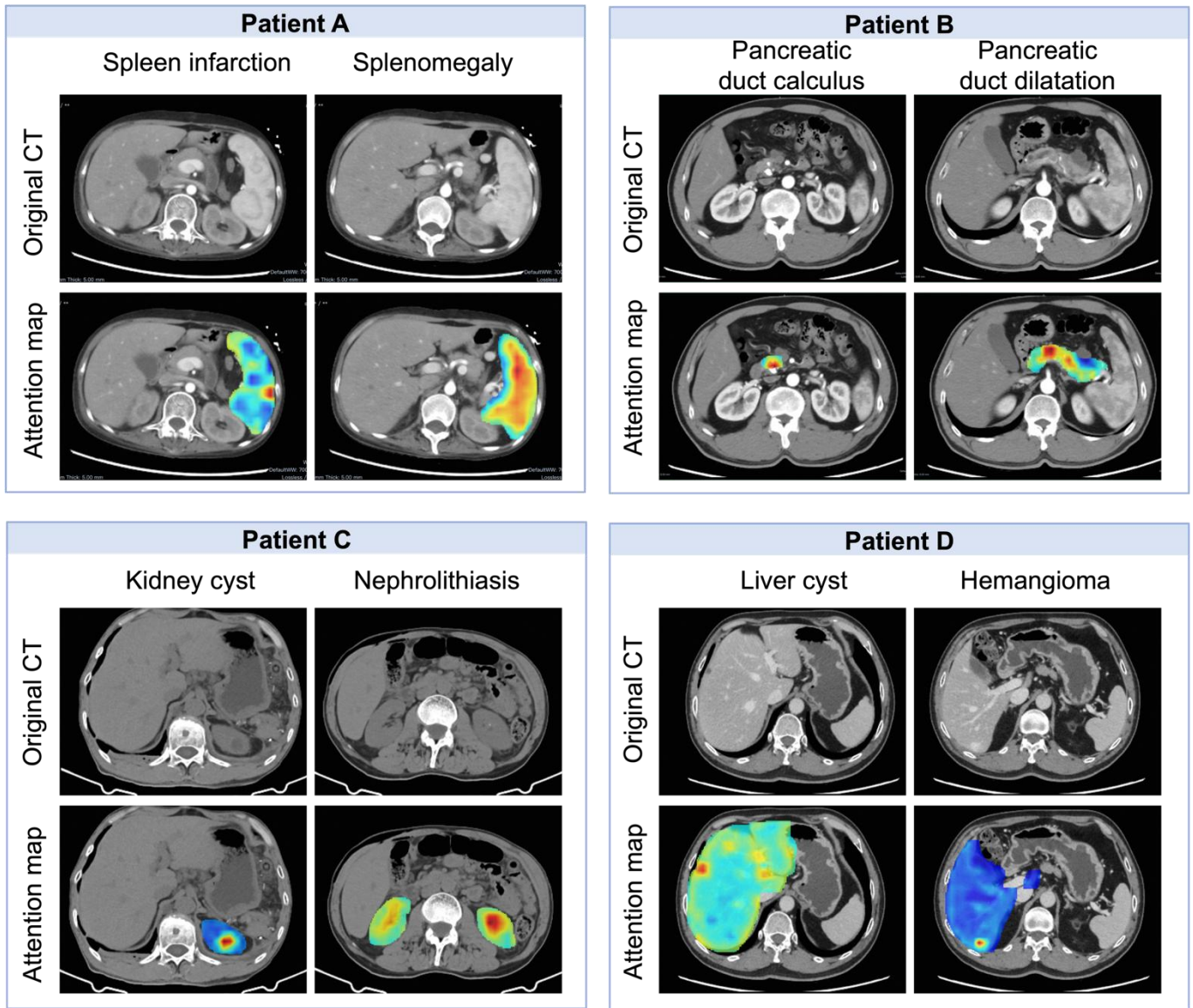


Figure S5. Disease-specific Grad-CAM visualizations for different pathologies affecting the same organ. Four patient cases (Patient A–D) are presented for the spleen, pancreas, kidney, and liver. For each patient, the upper row shows the original CT image and the lower row shows the corresponding Grad-CAM heatmap generated under a specific disease prompt. The examples illustrate that RADAR produces distinct spatial activation patterns for different diseases within the same organ, rather than uniformly highlighting the most visually conspicuous abnormal region.

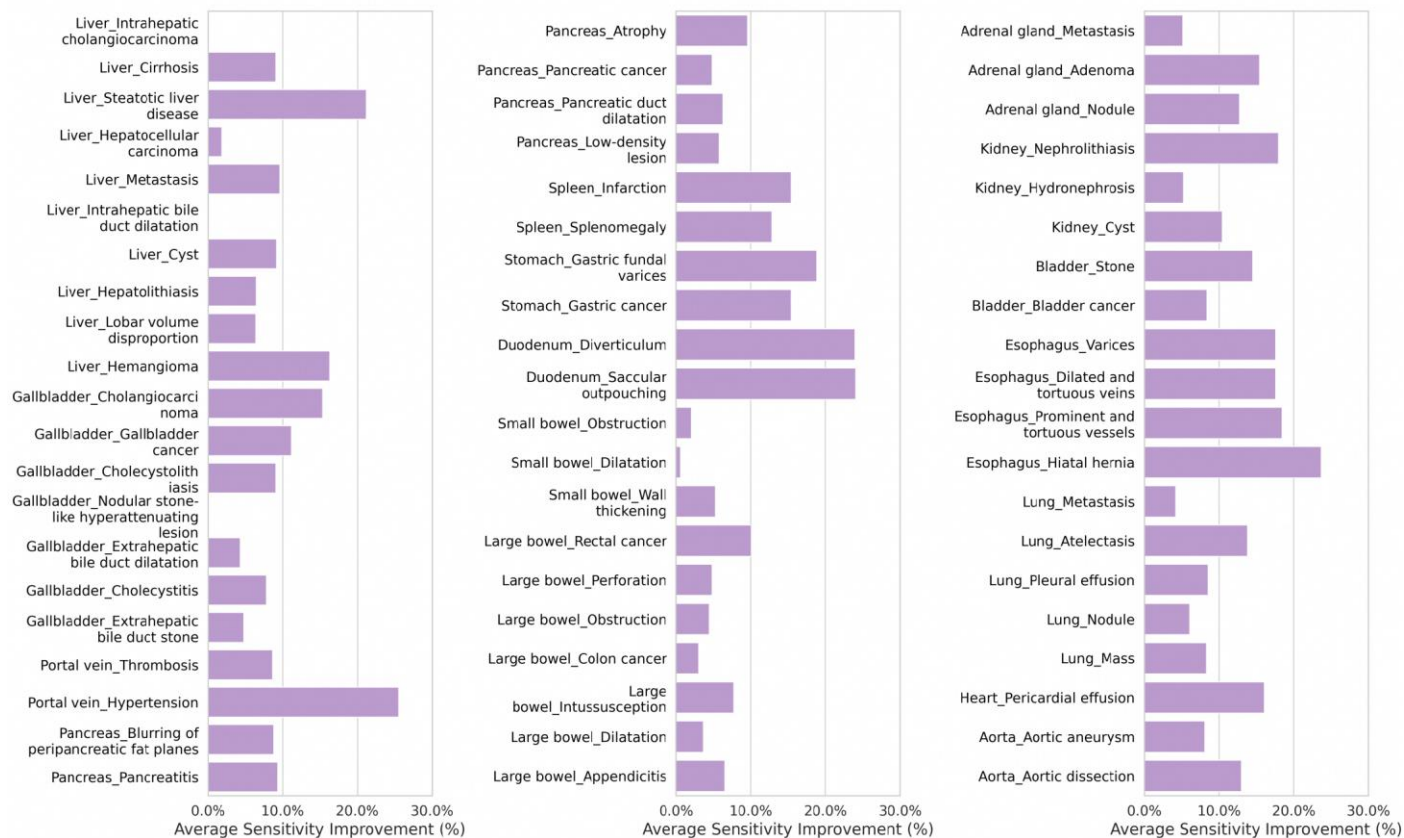


Figure S6. Average sensitivity improvement of different findings in reader studies. The bar chart displays the average improvement, averaged across all readers, in diagnostic sensitivity for each of the 61 findings evaluated in the second-session reader study. Each bar represents a single disease.

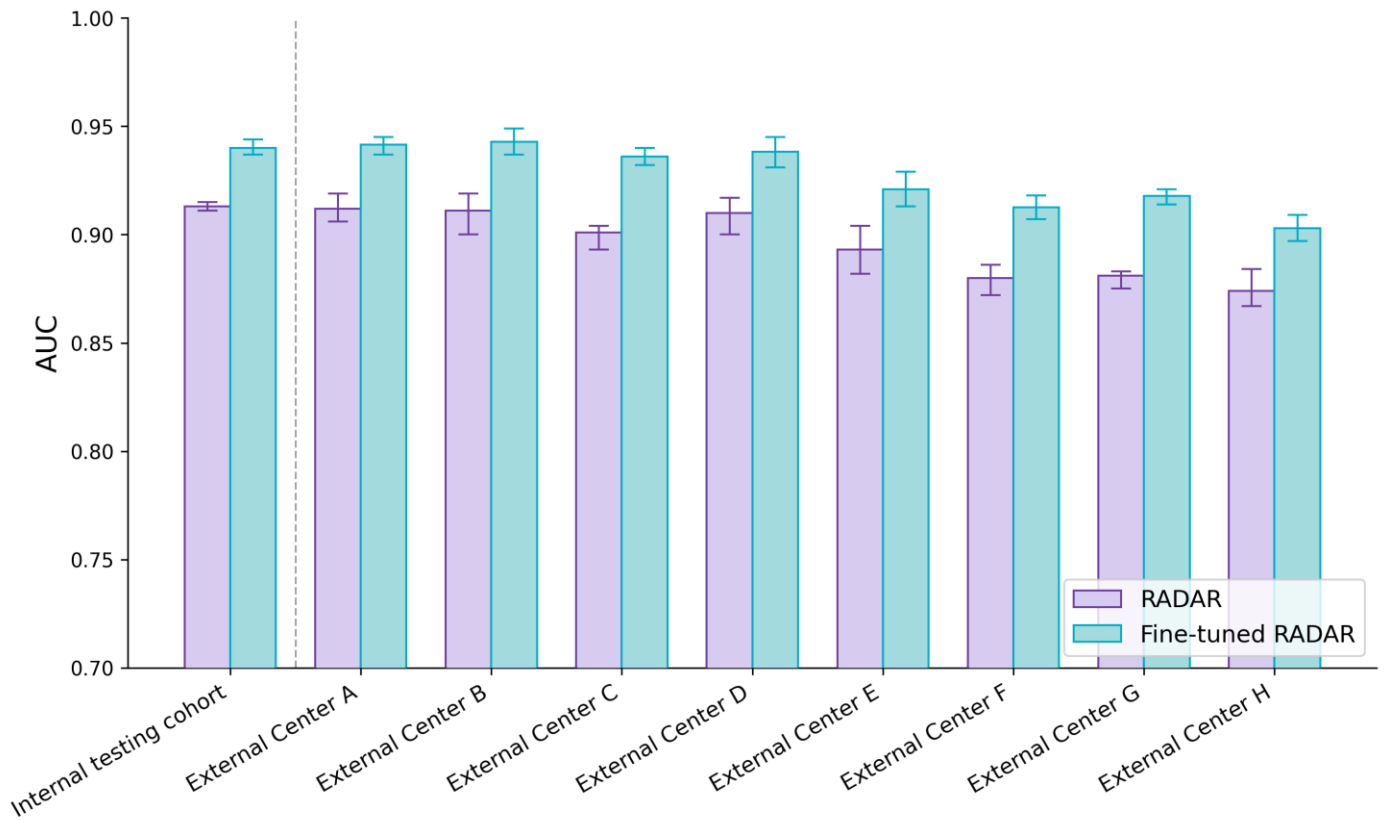


Figure S7. Comparison of RADAR and fine-tuned variant on internal and external test cohorts. Fine-tuned RADAR consistently achieves higher averaged AUC of 146 findings than the original RADAR across all centers. Error bars, 95% confidence intervals.



Figure S8. Data curation workflow for constructing RAD-CT. The RAD-CT dataset was built through a multi-stage pipeline. First, abdominal CT examinations were collected and de-identified, followed by contrast-phase identification. In total, the dataset contains 424,911 examinations and 1,497,673 volume-wise image–text pairs. A key innovation enabling fine-grained vision–language training was the refinement of the data from the whole-volume level to the anatomy level via CT anatomy parsing and anatomical decomposition of the

corresponding reports. After excluding non-intact organs, the final dataset comprised 15,523,242 anatomy-wise image–text pairs.

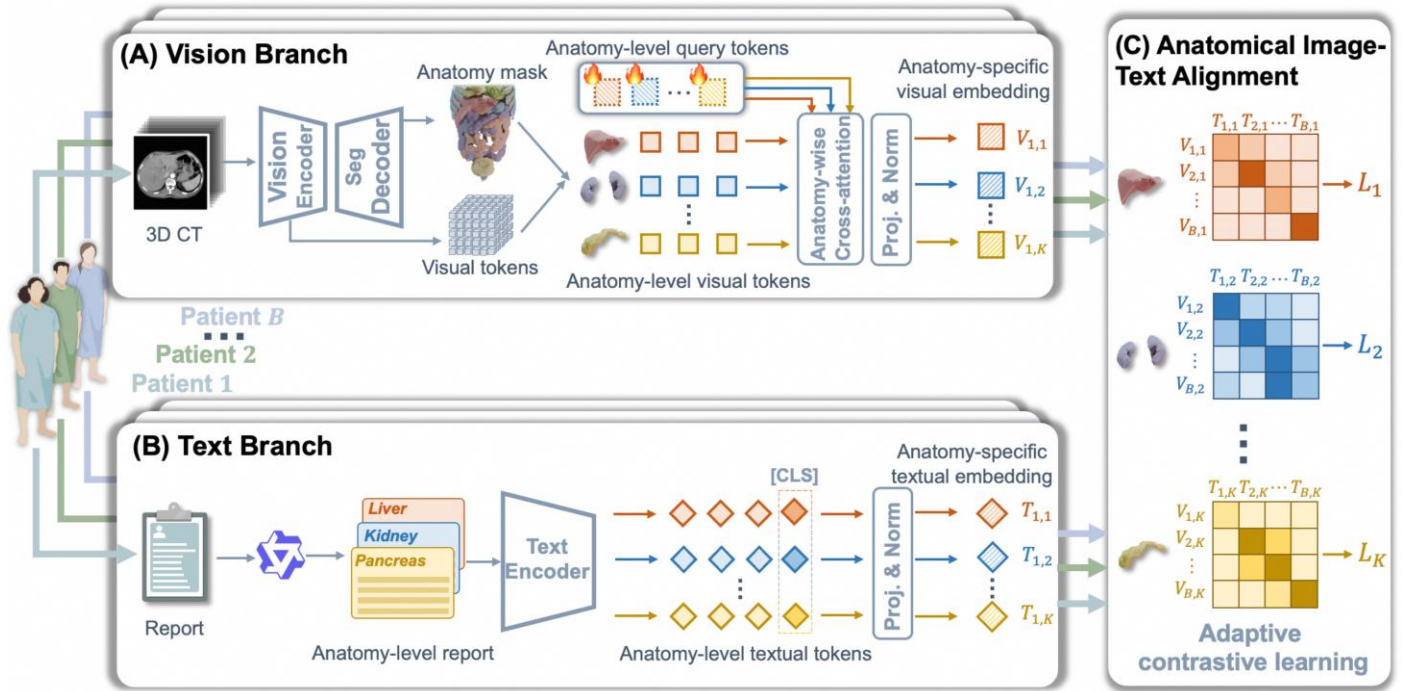


Figure S9. Overview of the RADAR architecture. (A) The vision branch uses a 3D vision encoder and an anatomical perception module to extract anatomy-specific visual features, aggregated via anatomy-wise cross-attention into visual CLS tokens. (B) The text branch tokenizes anatomy-specific report fragments and encodes them with a BERT-base text encoder to obtain text CLS tokens. (C) Anatomical image–text alignment is achieved through adaptive contrastive learning. [Figure created with Biorender.com]

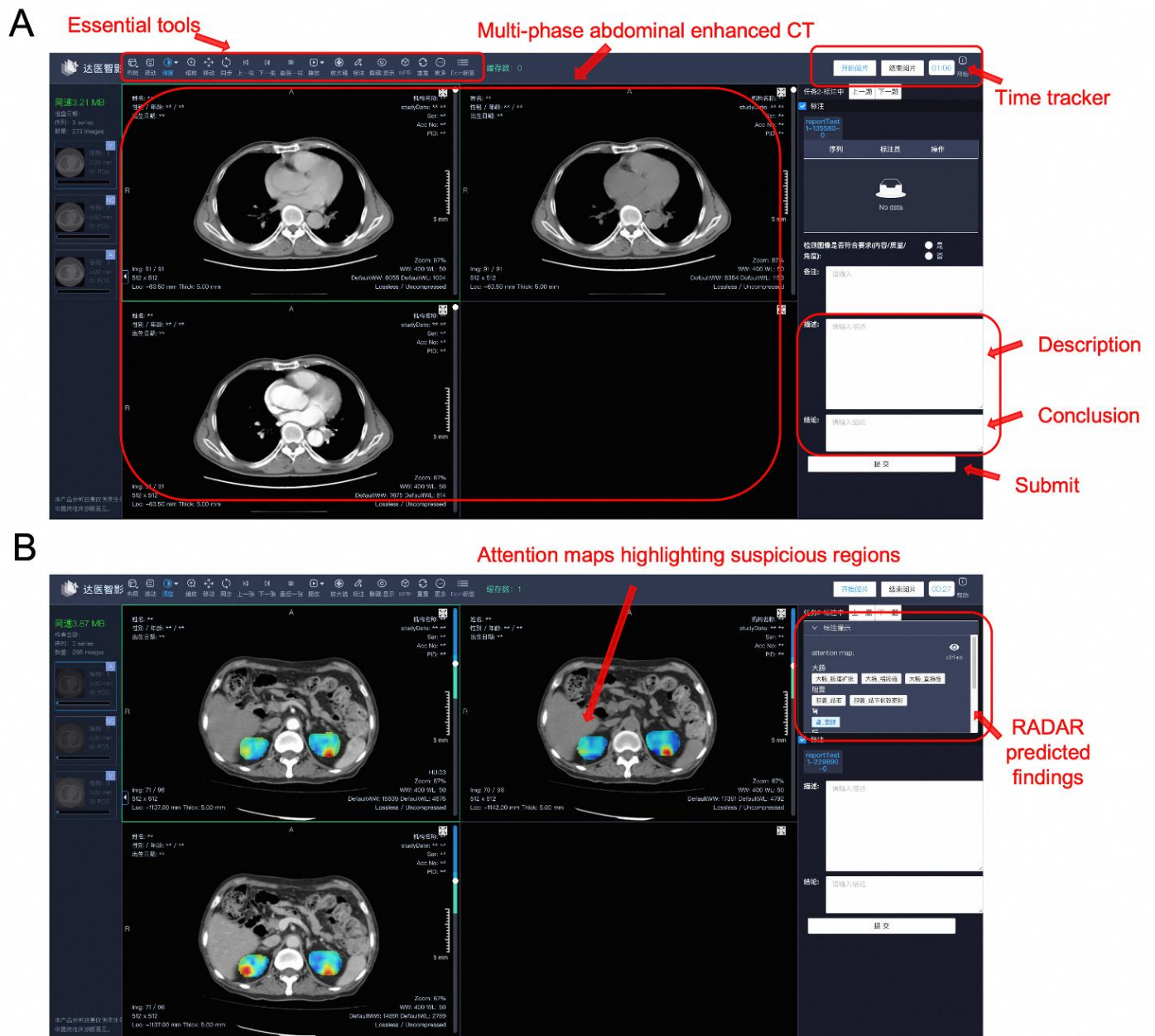


Figure S10. Reader study platform. (A) The user interface for the unaided reading session (first session). The platform displayed multi-phase CT scans (e.g., non-contrast, arterial, and venous phases) and provided essential radiological tools. Readers were instructed to complete dedicated fields for a “Description” and “Conclusion”, while a built-in timer recorded the reading duration for each case. (B) The user interface for the AI-assisted reading session (second session). In addition to the features shown in the first session, this version of the platform provided readers with an AI-generated list of suspected positive findings. For each finding on the list, a corresponding attention map was available, which

could be overlaid onto the CT images or hidden by the reader, to highlight the region of interest identified by the model.

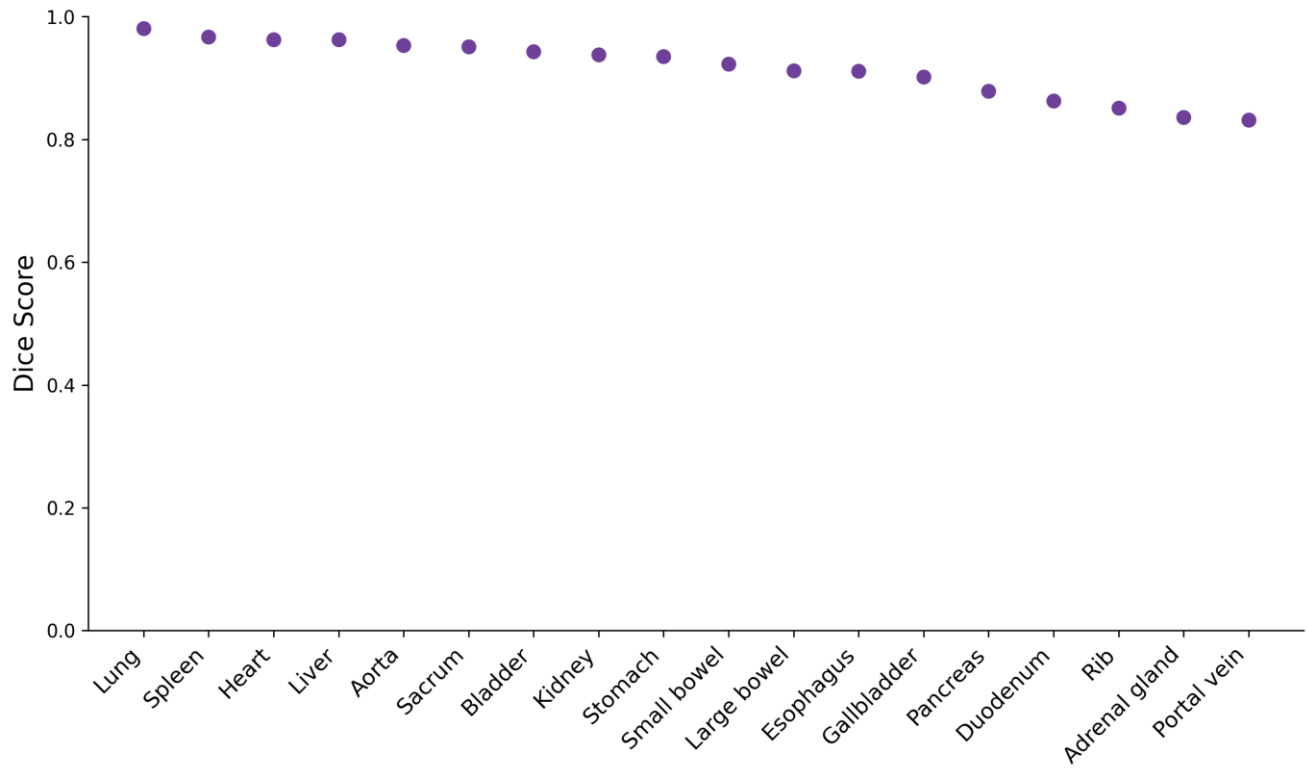


Figure S11. Evaluation of the anatomical perception module's segmentation performance. Each point represents the dice score for one of the 18 target anatomies on the TotalSegmentator test set.

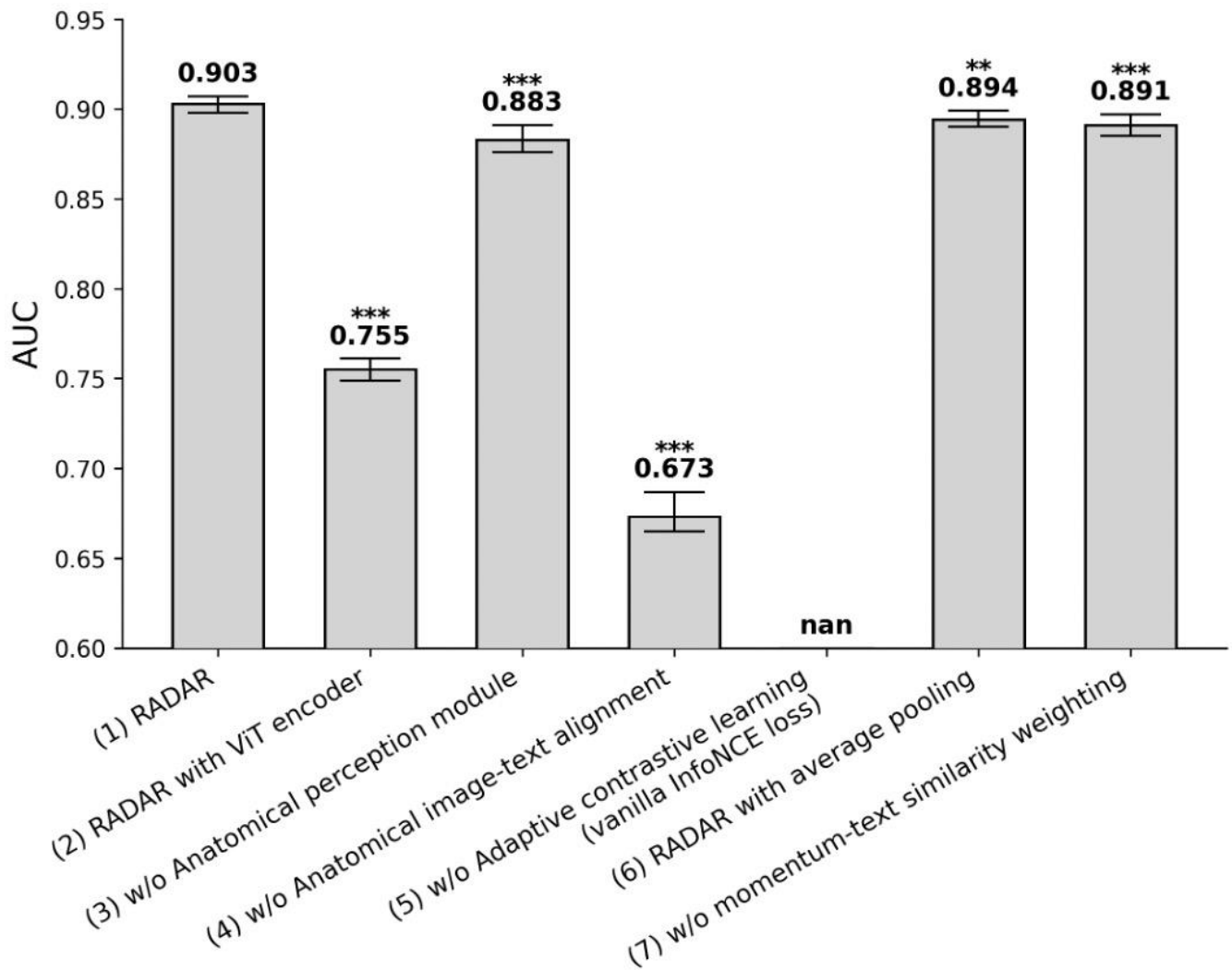


Figure S12. Ablation study of the RADAR framework. The chart illustrates the impact of different components on the diagnosis performance (AUC). These columns denote our full RADAR model and variants where specific modules are removed or replaced. These include: (1) our full RADAR model, (2) visual encoder (ViT vs. CNN backbone), (3) anatomical perception module, (4) image–text alignment strategy (whole-image vs. anatomical level), (5) contrastive loss (vanilla InfoNCE vs. adaptive contrastive loss), (6) feature aggregation strategies (average pooling vs. learnable query tokens), and (7) momentum-text similarity weighting. Error bars, 95% confidence intervals. Asterisks denote statistical significance: ** $P < 0.01$, and *** $P < 0.001$, compared to RADAR. nan, not a number (due to fail to converge of the model).

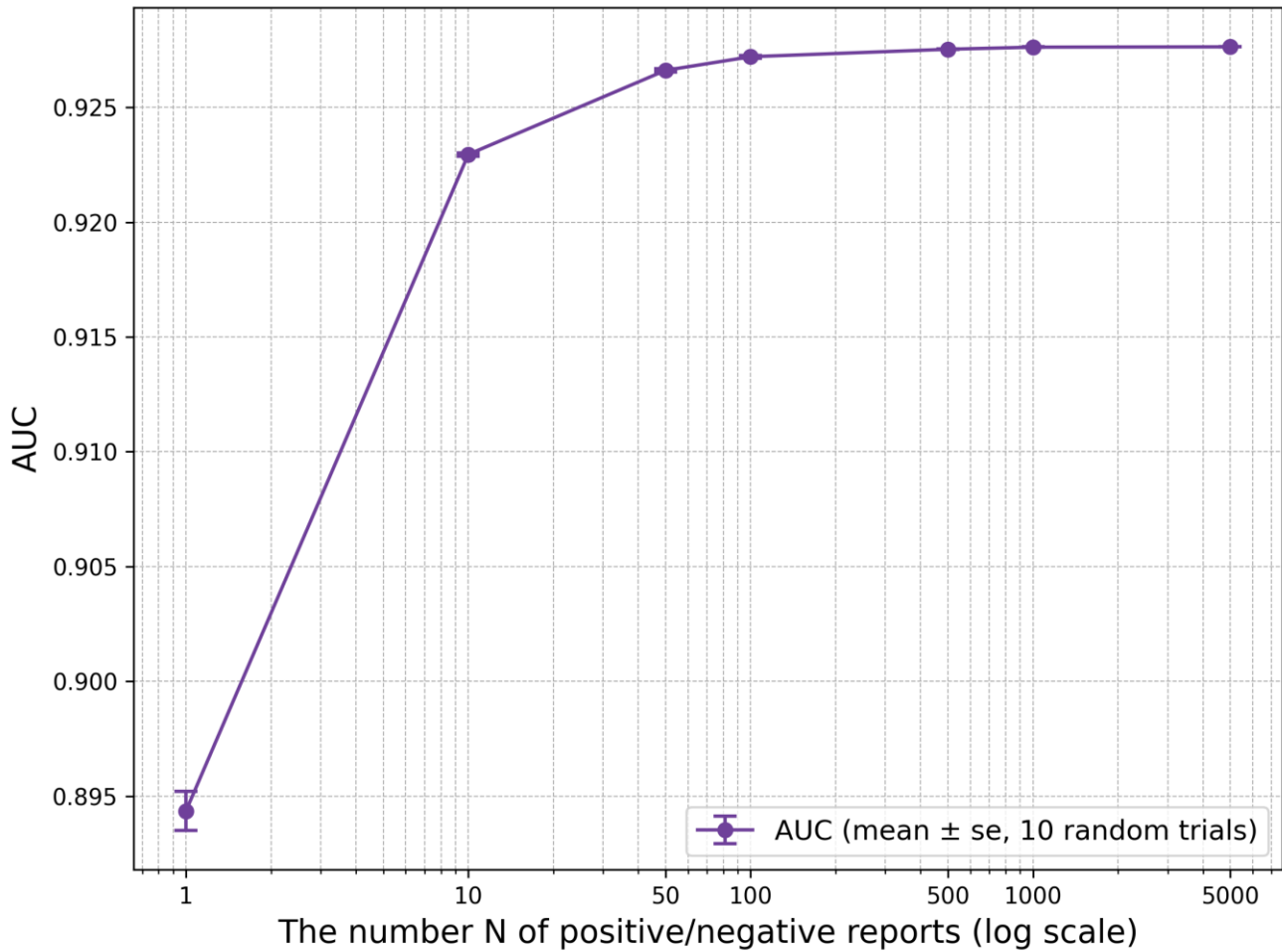


Figure S13. Effect of the number N of positive/negative reports on prompt ensemble performance. Each point shows the average AUC over 10 random trials, with error bars indicating the standard error. The X-axis is plotted on a logarithmic scale.

Table S1. Data distribution. Positive sample distributions for 146 findings across multiple testing cohorts of internal center, eight real-world external centers, as well as emergency and reader study datasets.

Table S2. Comparison of AUCs for CT-CLIP, Merlin, fVLM and RADAR across 18 anatomies and 146 findings in internal testing cohort. For each finding, the statistical

significance of the AUC difference between RADAR and each of the other three methods was assessed using DeLong’s test. Significance levels are denoted as follows: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Table S3. Assignment of findings to different functional systems.

Table S4. Categorization of findings based on hollow and solid organ types.

Table S5. Assignment of findings to signs and diseases.

Table S6. Assignment of disease findings to benign and malignant.

Table S7. AUC performances for 146 findings across eight real-world external centers.

Findings corresponding to the same anatomical region are represented using the same color. The number of findings for each center is as follows: Center A with 118 findings, Center B with 86 findings, Center C with 131 findings, Center D with 68 findings, Center E with 71 findings, Center F with 115 findings, Center G with 136 findings, and Center H with 93 findings.

Table S8. Detailed performances of the cross-population evaluation on the public Merlin-CT-test benchmark. RADAR+ represents the fine-tuned version of RADAR on the Merlin-CT-Train dataset.

Table S9. Reader information. Professional characteristics of the radiologists participating in the reader study, stratified by hospital tier and levels of radiologist experience. Readers were divided into four groups: (G1) senior radiologists from top-tier hospitals, (G2) junior radiologists from top-tier hospitals, (G3) senior radiologists from other hospitals, and (G4) junior radiologists from other hospitals. “Reader ID” denotes the unique identifier for each participant within their group. “Experience (year)” indicates the total number of years in radiology practice. “Amount of average abdominal CT read per year” refers to the number of abdominal CT examinations interpreted annually in recent years.

Table S10. The 104 substructures in the TotalSegmentator dataset were grouped into 36 major anatomical structures. Note that anatomies that do not appear in abdominal CT, such as the brain and face, were excluded, as well as those with extremely low prevalence in both our internal real-world test set and disease cases, such as the gluteus and iliopsoas. Ultimately, 18 anatomies were selected as the target anatomies for RADAR and used for both training and evaluation.

Table S11. Detailed performances of the fine-tuned Qwen model in extracting 146 findings.

Table S12. Detailed performances of the queried Qwen2.5-max model in extracting 146 findings.

Table S13. Comparison of global-window and sliding-window inference in terms of AUC, speed, and GPU memory.